

Package ‘manMetaVAR’

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Title Multivariate Meta-Analysis of Vector Autoregressive Model
Coefficients

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coef.manmetavar.dtvvar *Parameter Estimates (FitDTVAR)*

Description

Parameter Estimates (FitDTVAR)
 Parameter Estimates (FitMetaVAR)
 Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
```

```

    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, ...)

```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
vanishing_theta	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
theta_tol	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
...	additional arguments.
median	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.

Value

Returns a list of vectors of parameter estimates.

Returns a vector of estimated parameters.

Returns a vector of estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

Compress

Compress Replication

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

taskid Positive integer. Task ID.
 repid Positive integer. Replication ID.
 output_folder Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

confint.manmetavar.metavar

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Description

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Confidence Intervals for the Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, ...)
```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
lb	Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.
...	additional arguments.

Value

Returns a matrix of confidence intervals.
Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Dynamics	<i>Process Dynamics</i>
----------	-------------------------

Description

Process Dynamics

Usage

`Dynamics(dynamics)`

Arguments

dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
----------	--

See Also

Other Data Generation Functions: [GenData\(\)](#)

Examples

```
## Not run:
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)

## End(Not run)
```

FitDTVAR

*Fit the Model using the fitDTVARMxID Package***Description**

The function fits the model using the [fitDTVARMxID::fitDTVARMxID](#) package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

Description

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL)
```

Arguments

<code>fit</code>	R object. Output of the FitDTVAR() function.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

FitMLVAR

Fit the Model using the mlVAR Package

Description

The function fits the model using the [mlVAR::mlVAR](#) package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)

## End(Not run)
```

FitMplus

Fit the Model using Mplus

Description

The function fits the model using Mplus.

Usage

```
FitMplus(data, wd = ".", mplus_bin = NULL, ncores = NULL)
```


Arguments

data	R object. Output of the GenData() function.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

GenData

Simulate Data

Description

The function simulates data using the [simStateSpace::SimSSMIVary\(\)](#) function.

Usage

```
GenData(taskid)
```

Arguments

taskid	Positive integer. Task ID.
--------	----------------------------

See Also

Other Data Generation Functions: [Dynamics\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)

## End(Not run)
```

MplusInput

Create Mplus Input File

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data = NULL,
  fn_estimates = NULL,
  fn_results = NULL,
  fn_posterior = NULL,
  fn_factorscores = NULL,
  ncores = NULL
)
```

Arguments

fn_data	Character string. Filename for data file.
fn_estimates	Character string. Filename for estimates output.
fn_results	Character string. Filename for results output.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

Examples

```
cat(MplusInput())
```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

params

Format

A dataframe with 25 rows and 3 columns:

taskid Simulation Task ID.**n** Sample size.**time** Number of measurement occasions.**dynamics** Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.**Author(s)**

Ivan Jacob Agaloos Pesigan

plot.manmetavar.data	<i>Plot Method for an Object of Class manmetavar.data</i>
----------------------	---

Description

Plot Method for an Object of Class manmetavar.data

Usage

```
## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

x	Object of class manmetavar.data.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.data *Print Method for an Object of Class manmetavar.data*

Description

Print Method for an Object of Class manmetavar.data

Print Method (FitDTVAR)

Print Method (FitMetaVAR)

Print Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.dtvvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

x	Object of class <code>manmetavar.mlvar</code> .
...	additional arguments.
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
alpha	Numeric vector. Significance level α .
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
vanishing_theta	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
theta_tol	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
digits	Integer indicating the number of decimal places to display.
show	Tables to show.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim	<i>Simulation Replication</i>
-----	-------------------------------

Description

Simulation Replication

Usage

```
Sim(taskid, repid, output_folder, overwrite, integrity, seed, mplus = FALSE)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
mplus	Logical. If mplus = TRUE, fit a DSEM model in Mplus.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR	<i>Simulation Replication - FitDTVAR</i>
-------------	--

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR

Simulation Replication - FitMetaVAR

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR	<i>Simulation Replication - FitMLVAR</i>
-------------	--

Description

Simulation Replication - FitMLVAR

Usage

SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus	<i>Simulation Replication - FitMplus</i>
-------------	--

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .

Value

Returns a character string file name with the `output_folder` in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj	<i>Simulation Project Name</i>
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Description

Simulation Project Name

Usage

SimProj()

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	<i>Summary</i>
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Description

Summary

Usage

Sum(taskid, reps, output_folder, overwrite, integrity, mplus = FALSE)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
mplus	Logical. If mplus = TRUE, fit a DSEM model in Mplus.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARLB

Summary: Likelihood-Based (FitMetaVAR)

Description

Summary: Likelihood-Based (FitMetaVAR)

Usage

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal	<i>Summary: Normal Theory (FitMetaVAR)</i>
---------------------	--

Description

Summary: Normal Theory (FitMetaVAR)

Usage

SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMLVAR	<i>Summary (FitMLVAR)</i>
Description	
Summary (FitMLVAR)	
Usage	
SumFitMLVAR(taskid, reps, output_folder, overwrite, integrity)	
Arguments	
taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
Details	
This function is executed via the Sum function.	
Value	
The output is saved as an external file in output_folder.	
Author(s)	
Ivan Jacob Agaloos Pesigan	

SumFitMplus	<i>Summary (FitMplus)</i>
Description	
Summary (FitMplus)	
Usage	
SumFitMplus(taskid, reps, output_folder, overwrite, integrity)	

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar.data

Summary Method for an Object of Class manmetavar.data

Description

Summary Method for an Object of Class manmetavar.data

Summary Method (FitDTVAR)

Summary Method (FitMetaVAR)

Summary Method (FitMLVAR)

Summary Method (FitMplus)

Usage

```
## S3 method for class 'manmetavar.data'
summary(object, ...)
```

```
## S3 method for class 'manmetavar.dtvvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
```

```

    psi = TRUE,
    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    digits = 4,
    ...
)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, ...)

```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
...	additional arguments.
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
alpha	Numeric vector. Significance level α .
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .

vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
digits	Integer indicating the number of decimal places to display.
show	Tables to show.
median	Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, number of posterior samples, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

vcov.manmetavar.dtvvar *Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)*

Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```

    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, ...)

```

Arguments

<code>object</code>	Object of class <code>manmetavar.mplus</code> .
<code>alpha</code>	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
<code>beta</code>	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
<code>nu</code>	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
<code>psi</code>	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
<code>theta</code>	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
<code>converged</code>	Logical. Only include converged cases.
<code>grad_tol</code>	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
<code>hess_tol</code>	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
<code>vanishing_theta</code>	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
<code>theta_tol</code>	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
<code>...</code>	additional arguments.

Value

Returns a list of sampling variance-covariance matrices.

Returns the sampling variance-covariance matrix of the estimated parameters.

Returns the sampling variance-covariance matrix of the estimated parameters.

Author(s)

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